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Phase Prediction via Crystal Structure Similarity in the Periodic Number Representation

Cem Oran, Riccarda Caputo,* Pierre Villars,* Hasan Bilal Özcü, Feraye Hatice Canbaz, and Adem Tekin*

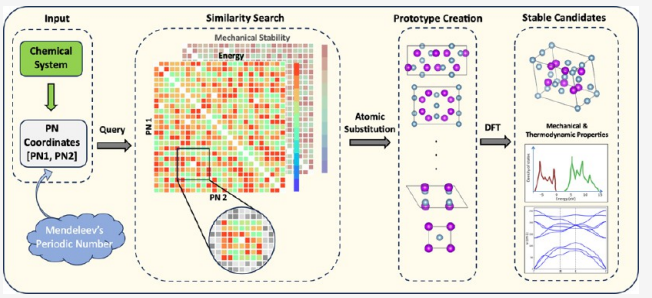
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ABSTRACT: The periodic number (PN) representation of the chemical systems, introduced by Dmitri Mendeleev, uncovers the fundamental principle of chemical similarity in a straightforward way. In this framework, the rows correspond to the principal quantum numbers of the elements' electronic configurations when considered isolated atoms. This systematic arrangement allows for a deeper understanding of the relationships and patterns among the elements. In this study, we propose a novel strategy for structure type (prototype) prediction by utilizing the PN concept to identify possible modifications and phase stability of unexplored chemical systems. Our PN-based crystal structure prediction (PNcsp) program, which evaluates similarity through PN neighboring in the phase map, provides the most probable prototypes for unknown and unreported modifications of given phases in binary and higher order chemical systems. We applied PNcsp to 59 distinct chemical systems whose equimolar phases are indicated in the respective phase diagrams but lack accurate experimental structure determination. Our methodology identified 93 prototypes for these 59 equiatomic phases, of which 47 exhibit mechanical and dynamic stability. Notably, this approach discovered 19 entirely novel, fully stable polymorphic phases, thereby expanding the known landscape of potential materials. Furthermore, we demonstrated that this method is also effective for nonequimolar and higher order systems.



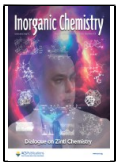
INTRODUCTION

The potential of contemporary computational materials science is captivating, particularly its ability to predict crystalline properties of compounds not yet observed experimentally. Recent advancements in the theoretical and computational methods have facilitated the systematic investigation of a large class of compounds.[1−7] This investigation is driven by the demand for discovering novel materials with finely tuned properties, thus catalyzing innovation across diverse technological applications ranging from electronics to energy storage.

Additionally, the integration of machine learning (ML) algorithms and high-performance computing platforms has further augmented the predictive capabilities of computational materials science, enabling more accurate and expedited analyses of a large number of new materials.[8−14] While ML models demonstrate notable success in predicting material properties and structure types (prototypes), significant drawbacks persist. These include their requirement on large data sets and limited interpretability that hinders the extraction of meaningful chemical insights.

An alternative approach is based on the direct application of theoretical and empirical chemical knowledge to formulate predictions about the crystal structures of novel compounds by using ab initio global optimization techniques. However, this approach relies heavily on computationally intensive density functional theory (DFT) calculations, which restricts its applicability only to relatively small chemical systems. At present, a commonly utilized strategy in the prediction of crystalline properties for unknown chemical systems involves scanning the composition space while adhering to fixed crystal structure prototypes.[15−19] This strategy is particularly advantageous because it capitalizes on well-established crystallographic data, reducing the computational complexity associated with structural optimization. Additionally, different authors[18−20] have demonstrated the efficiency of exploring the space of potential stable materials through assessments of chemical similarity among constituent elements. Building upon this approach, we introduce a novel likelihood metric based on Mendeleev Periodic Number (PN), which was extensively

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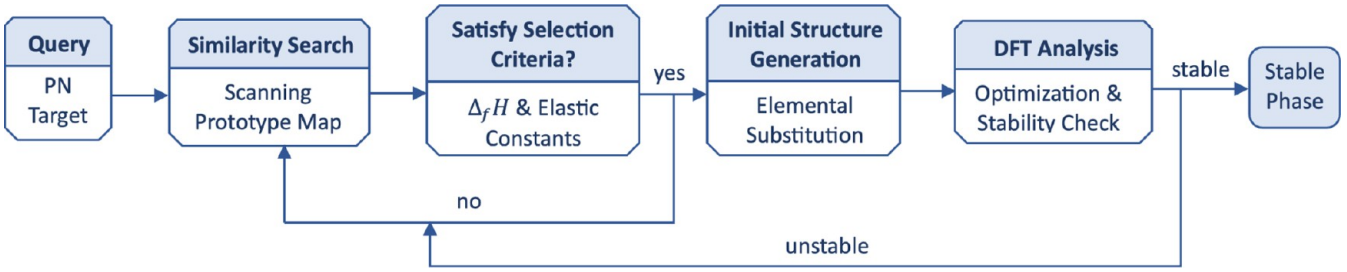


Figure 1. Workflow of PNcsp. The program predicts stable phases based on the crystal structure type similarity in the PN representation of the phase map. Stable phases then undergo further analyses for the investigation of the electronic properties.

detailed in the Supporting Information (SI) and in our previous works.[21,22]

Our methodology is based on transforming established chemical systems by replacing atoms within their crystal structures with others possessing similar chemical properties. The existence of similarity is quantitatively defined by extracting a measure of chemical interchangeability from the PN enumeration which represents a significant distinction from the similar methods reported in prior studies by other groups.[19] To enhance the selection of stable prototype candidates, we supplement thermodynamic stability criteria with an additional measure of mechanical stability. Subsequent in-depth analysis of the mechanical and dynamical stability of our proposed candidates reveals the substantial predictive accuracy of this approach. The identification of the stable phases in the PN representation of the phase map makes it possible to streamline the selection process and to reduce considerably the size of the data set to analyze.

As reported in detail,[22] the PN enumeration of the chemical elements enables us to integrate the criteria of principal quantum number with the valence shell configuration to distinguish the rows and groups in the Periodic Table. Moreover, PN enumeration leads us to create a phase map that distinctly emphasizes areas in inorganic systems where phases form (formers) and do not form (nonformers). This phase map corresponds to a 2-dimensional map for binary systems, which is constructed by using the PNs of the constituent chemical elements of the chemical systems in increasing order. The scope of PN ranges separating nonformer and former regions in binary systems can be expanded to encompass ternary chemical systems and beyond, simply by leveraging the PN of the constituent chemical elements. Even though the appearance of the phase map dates back to 80s, the notable difference of the present representation is its ability to clearly indicate regions where the phases of the given chemical system form, hence former regions, and regions where phases do not form, nonformer regions.

In addition, the presented phase map showcases trends and similarities in the chemical properties of neighboring chemical systems, such that computed enthalpy of formation, mechanical properties and prototypes, and effectively reveals patterns and resemblances among phases formed by chemical elements sharing the same periodic groups. For instance, similarities can be observed in the equiatomic binary combination of Cu with the complete range of rare earth (RE) elements.[21] A similar line of reasoning can be applied to a large variety of chemical groups, such as Ag-RE and Au-RE systems and in the prototype of the monopnictides and monohydrides.

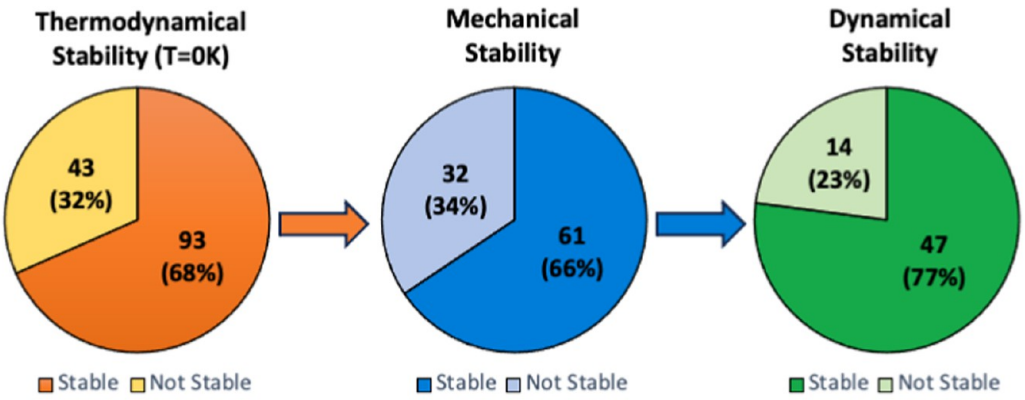
The comprehensive perspective emphasized by the PN representation also possesses predictive potential when it comes to seeking yet-to-be-discovered or unreported chemical systems. Certainly, the crystal structures of thermodynamically and mechanically stable phases at particular compositions can be utilized to forecast the initial structure prototypes of undiscovered systems composed of chemical elements with PNs similar to those in well-known systems. In this paper, we introduce our PN-based crystal structure prediction (PNcsp) algorithm, aimed at uncovering potential equiatomic binary phases that have not yet been reported. In addition, to test the prediction accuracy of our method we picked up a few examples beyond the equimolar binary phases, such as nonequimolar binary, ternary and quaternary systems. This generalized version of the program is publicly available on Github.

METHODOLOGY

Crystal Structure Similarity Approach. An exhaustive examination of the entire LINUS PAULING FILE Database (LPF)[23] (see also https://mpds.io/), including binary, ternary, and higher order chemical systems has been presented in our recent work.[22] In that study, an extensive collection of experimental data in LPF has been empowered with the first-principles calculations covering the entire space of equimolar binary systems in addition to the accepted violations falling in the first identified forbidden region (where binary phases do not form). The inspection of the available binary phase diagrams indicates that there are 936 chemical systems which exhibit the equimolar phase. Among them, there are 90 systems for which the crystallographic data of the equimolar phase has not yet been reported. The lack of available experimental structure data of such phases prompted us to explore the structure and phase stability by employing a first-principles crystal structure prediction approach. Therefore, in the present work, we focus on these systems and in particular on the ones which are formed by the elements we can treat computationally (in the complete set of calculations, we considered 77 out of 118 chemical elements that were reported in[22]). In total, there are 59 distinct chemical systems whose equimolar phases are indicated in the corresponding phase diagrams, but their crystal structures have not yet been reported. For these systems, initial prototype predictions were obtained by applying the structure type similarity concept from neighboring systems within the PN representation. In fact, stable phases of chemical systems close in the PN representation most likely adopt similar structure types.

Using the PN representation in a discrete two-dimensional map, within the first nearest neighbor region, a given system can have up to 8 possible neighbors, each having at least one

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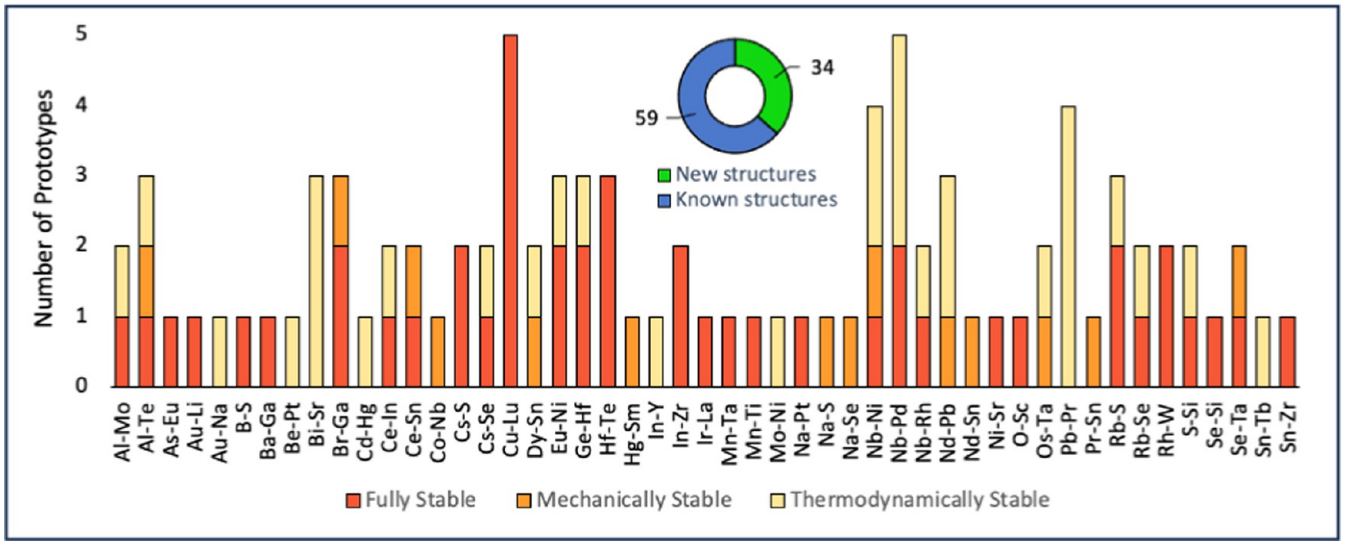


Figure 2. Distribution of 136 structures predicted via similarity method. **Up:** From left to right, the number of predicted stable structures in terms of enthalpy of formation, mechanical stability and dynamical stability are presented. **Bottom**: Distribution of the number of predicted structures among distinct chemical systems. Colors distinguish fully stable structures from those satisfying only mechanical or thermodynamic stability conditions. Among these total 93 structures, 34 of them are novel and proposed only by this study. The remaining 59 were previously proposed in various studies.

modification. We narrowed the possibilities by selecting only the neighbors with thermodynamic and mechanical stability. Consequently, for the prototype selection process, we developed PNcsp algorithm employing 2 levels of selection criteria for the neighboring phase: the enthalpy of formation, which must be negative, and the mechanical stability, which ensures that no structural phase transformation occurs upon full geometry optimization. The maps of the enthalpies of formation and the mechanical stability of the equimolar phases, already reported in ref 22, were the starting point for the search algorithm.

As summarized in Figure 1, the algorithm starts the investigation with the closest neighbors (first order) and moves on to the second and third-order neighbors. The prototypes satisfying the criteria are selected for the atomic substitution step. We limited the neighbor search up to three neighbors for each constituent element in the binary. This means that the algorithm moves through the phase map in each direction at most three times and chooses the best prototype candidates. Therefore, a binary chemical system has 6 neighboring elements for each of its constituent elements. For each of these neighbor phases, priority is given to the room-temperature modification, and, if not present, to other possible stable modifications, either low- or high-temperature. In this way, the most likely prototype candidates in the neighborhood were selected for further analysis. For the selected prototypes, first-principles total energy calculations were employed to determine their thermodynamic properties, specifically the enthalpies of formation and cohesive energies, as well as their mechanical properties, including elastic constants and bulk modulus. Furthermore, phonon dispersion calculations were conducted on the selected structures to ascertain their dynamic stability and deduce the associated thermodynamic functions. Phonon calculations were conducted only for prototypes that are thermodynamically and mechanically stable.

Computational Settings. For all calculations, we employed DFT using the CASTEP code[24] with the generalized gradient approximation (GGA) of the exchange-correlation functional of the PBEsol type.[25] Optimization of structures in terms of atomic positions and cell parameters was conducted using the BFGS algorithm.[26] We utilized the most recent set of on-the-fly generated pseudopotentials, namely the 80.otfg type, which have been demonstrated to minimize errors compared to all-electron DFT calculations.[27]

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Table 1. Structure Type, Enthalpy of Formation at T = 0 K, Gibbs Free Energy of Formation at T = 298 K, Constant Volume Heat Capacity, and Bulk Modulus of the 47 Equimolar Phases in Their Stable Modifications, All of Which Fulfill the Mechanical and Dynamical Stability Criteria[a]

chemical system	structure type	$\Delta_f H(0)$ [kJ/mol]	$\Delta_f G$ (298) [kJ/mol]	C_v (298) [J/K mol]	bulk modulus [GPa]
Al−Mo	**FeB-b,oP8,62**	**−35.88**	**−42.58**	**4.42**	**194.64**
Al−Te	SiAs,mS24,12	−60.84	−72.82	4.59	19.96
Al−Te	**GaS-a,hP8,194**	**−59.68**	**−71.95**	**4.58**	**22.95**
As−Eu	NaCl,cF8,225	−207.66	−217.86	4.86	52.64
Au−Li	CsCl,cP2,221	−83.69	−94.42	4.34	76.86
B−S	**GaS-a,hP8,194**	**−75.49**	**−78.24**	**3.13**	**84.38**
Ba−Ga	NaHg,oS16,63	−70.60	−87.70	4.84	24.72
Br−Ga	**TlI,oS8,63**	**−121.89**	**−142.17**	**4.94**	**25.44**
Br−Ga	NaCl,cF8,225	−113.27	−133.03	4.94	23.89
Ce−In	CsCl,cP2,221	−65.18	−89.41	4.89	57.42
CeSn	TlI,oS8,63	−120.30	−131.67	4.88	59.92
Cs−S	**CsTe,oP8,55**	**−176.69**	**−193.13**	**4.76**	**14.49**
Cs−S	NaCl,cF8,225	−61.63	−80.54	4.71	11.89
Cs−Se	**CsTe,oP8,55**	**−186.17**	**−205.44**	**4.90**	**14.73**
Cu−Lu	**FeB-b,oP8,62**	**−47.28**	**−60.96**	**4.85**	**78.95**
Cu−Lu	CuTi,tP2,123	−49.00	−63.50	4.87	87.23
Cu−Lu	**CuTi,tP4,129**	**−19.76**	**−34.58**	**4.87**	**43.99**
Cu−Lu	**TiNi,mP4,11**	**−21.40**	**−37.77**	**4.90**	**55.13**
Cu−Lu	CsCl,cP2,221	−49.78	−64.65	4.88	87.29
Eu−Ni	**TlI,oS8,63**	**−30.82**	**−45.51**	**4.85**	**43.77**
Eu−Ni	CsCl,cP2,221	−1.88	−18.74	4.90	58.15
Ge−Hf	FeB-b,oP8,62	−140.19	−150.70	4.72	140.39
Ge−Hf	NaCl,cF8,225	−124.43	−135.15	4.72	124.83
Hf−Te	WC,hP2,187	−141.58	−153.29	4.80	116.61
Hf−Te	**FeAs,oP8,62**	**−136.34**	**−148.96**	**4.83**	**104.95**
Hf−Te	**NiAs,hP4,194**	**−126.40**	**−139.02**	**4.82**	**113.65**
In−Zr	**MoB,tI16,141**	**−47.53**	**−57.82**	**4.75**	**102.21**
In−Zr	CuAu,tP2,123	−75.79	−86.20	4.75	104.68
Ir−La	CsCl,cP2,221	−51.67	−68.64	4.93	95.90
Mn−Ta	CsCl,cP2,221	−44.11	−51.99	4.58	262.20
Mn−Ti	CsCl,cP2,221	−44.80	−51.42	4.51	201.94
Na−Pt	LiRh,hP2,187	−21.91	−33.01	4.74	112.64
Nb−Ni	CuAu,tP2,123	−17.91	−27.62	4.72	182.27
Nb−Pd	CuAu,tP2,123	−35.05	−45.57	4.76	193.11
Nb−Pd	**AuCd,oP4,51**	**−26.72**	**−37.40**	**4.77**	**185.02**
Nb−Rh	CuAu,tP2,123	−82.23	−93.41	4.76	243.66
Ni−Sr	TlI,oS8,63	−1.41	−14.55	4.81	33.01
O−Sc	NaCl,cF8,225	−614.89	−608.03	4.04	192.70
Rb−S	**CsTe,oP8,55**	**−164.42**	**−178.92**	**4.75**	**17.33**
Rb−S	**CsSb,mP32,14**	**−163.79**	**−178.61**	**4.75**	**16.70**
Rb−Se	**CsTe,oP8,55**	**−173.24**	**−190.82**	**4.89**	**16.54**
Rh−W	CuAu,tP2,123	−35.69	−44.99	4.71	323.89
Rh−W	AuCd,oP4,51	−36.13	−46.22	4.74	326.02
S−Si	**GeS,oP8,62**	**−50.88**	**−57.39**	**4.32**	**46.18**
Se−Si	**GeS,oP8,62**	**−39.66**	**−48.79**	**4.53**	**41.51**
Se−Ta	WC,hP2,187	−68.54	−81.35	4.66	179.62
Sn−Zr	FeB-b,oP8,62	−105.42	−111.24	4.75	109.11

[a]Bold entries indicate novel polymorphic phases.

In the geometry optimizations, we employed the most stringent convergence criteria feasible, ensuring that the total energy, maximum atomic displacement, maximum atomic force, and maximum lattice stress were all maintained below 0.005 meV atom^{-1}, 0.0005 Å, 0.001 eV Å^{-1}, and 0.02 GPa, respectively. The SCF (Self-Consistent Field) threshold for assessing the convergence of an SCF step and the separation of the k-point mesh were established at 2. 10^{-7} eV atom^{-1} and below 0.028 Å^{-1}, respectively. The energy cutoff employed for the plane wave basis set was consistently selected to exceed the cutoff energy of the single atomic pseudopotential. Initial geometry optimizations were conducted with spin polarization for nearly all phases.

The computations of elastic constants were employed to investigate lattice stability and to infer mechanical properties of optimized structures, whose phase exhibits negative enthalpy of formation. The lattice stability was evaluated by employing the necessary and sufficient conditions.[28] For those showing lattice

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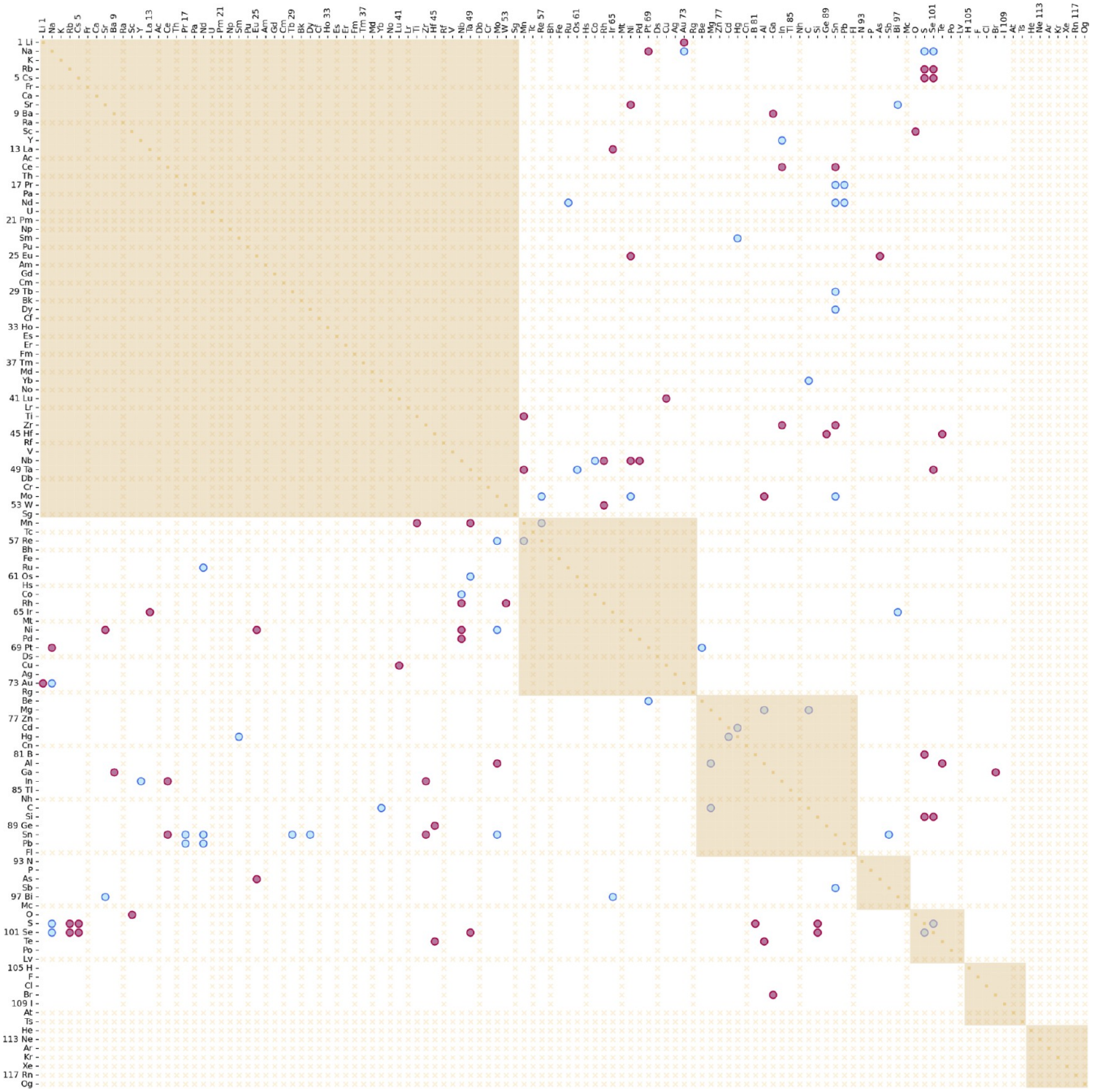


Figure 3. Locations of the 59 target equiatomic binary systems in the PN representation. Systems having at least one fully stable phase predicted via PN similarity are emphasized with red circles. The remaining systems are represented with blue circles. Pale orange-colored regions in the map indicate the seven forbidden regions where binary phases do not form, with the exception of a few allowed violations as discussed in detail in ref 22. This figure is adapted with permission from ref 22. Copyright [2024] [Elsevier].

stability, dynamical stability was also evaluated via phonon calculations. Both elastic constants and phonon calculations were conducted as implemented in the CASTEP code.

RESULTS AND DISCUSSION

For the 59 distinct chemical systems, whose equimolar phases are indicated in the respective phase diagrams, and whose crystallographic data are not yet available in the literature, at our best knowledge, our method successfully identified at least one stable modification for most of them. In fact, PNcsp's search algorithm proposed more than one possible structure type to start with the geometry optimization of the unknown phases. In total 136 distinct phases with different structure types were considered, among which 93 converged and the resulting enthalpies of formation were negative, hence indicating thermodynamic stability. For the thermodynamically stable phases, the mechanical stability was assessed via elastic constant calculations by using the necessary and sufficient conditions of the lattice stability,[28] which were fulfilled by 61 structures. Then for the mechanically stable structures, the phonon calculations helped to determine the dynamical stability. Ultimately, 47 phases exhibited both mechanical and dynamical stability, and we define them as fully stable equimolar phases. Figure 2 summarizes the outcomes of this ab

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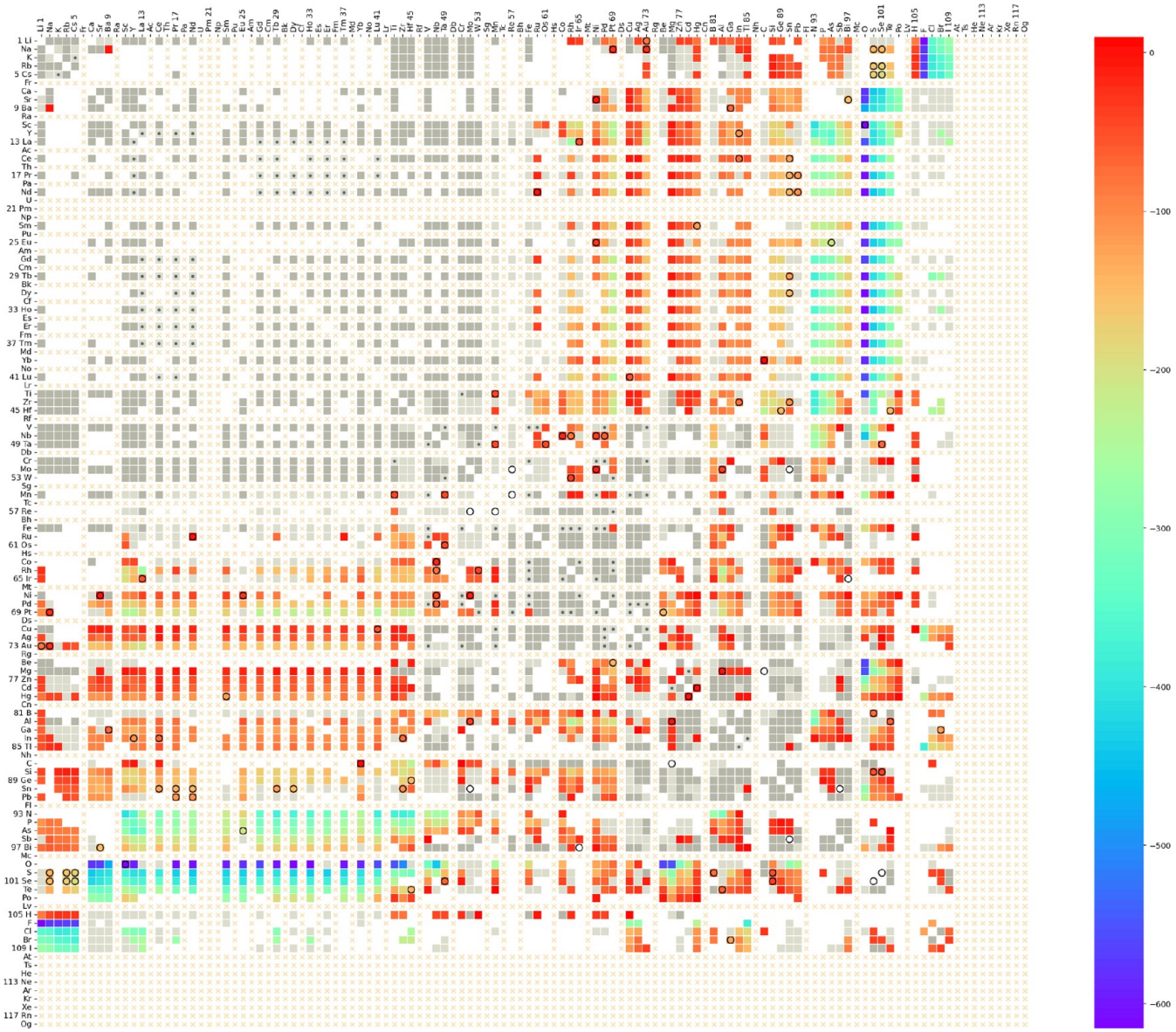


Figure 4. PN representation of the calculated enthalpies of formation (kJ/mol) of the new equiatomic binary phases overlaid on the values reported in ref 22. Negative values (indicating stable compounds) are in rainbow colors. Nonforming phases are dark gray, nonequiatomic formers are light gray, and ordered phases are light gray with dots. Excluded elements are marked with orange crosses. Cells with black empty circles (colors indicate predicted enthalpy of formation) correspond to the systems predicted via similarity. This figure is adapted with permission from ref 22. Copyright [2024] [Elsevier].

initio workflow. In Table 1, we report a list of the chemical systems with thermodynamic properties and bulk modulus of their fully stable modifications. We use the following format to describe the prototype: prototype compound, Pearson symbol and space group number, each separated by a comma (for example, the prototype of a rock-salt structure is NaCl,cF8,225). The Gibbs free energy of formation and the constant volume heat capacity are derived from the phonon dispersion calculations within the Debye model,[29] as implemented in the CASTEP code. Regarding mechanical properties, detailed results of the elastic constants calculations are provided in the SI.

Among the investigated systems, we found that the equimolar phase of the chemical systems Al−Mg, Bi−Ir, C−Mg, C−Yb, Mn−Re, Mo−Re, Mo−Sn, Nd−Ru, S−Se and Sb−Sn do not have any stable modification at T = 0 K. This somehow confirms the information we have from the analysis of the corresponding phase diagrams.[30] In fact, a strictly equimolar phase does not exist for any of them and most of the phases are represented either within the homogeneous composition range or at high temperatures. This observation also aligns with data from the Open Quantum Materials Database (OQMD),[3] where structures proposed for these systems also possess positive or near-zero enthalpies of formation. Notably, except Bi−Ir, C−Yb and Mo−Sn, these systems are located at the boundaries of forbidden regions in the PN representation of the binary phase map: Mn−Re, Mo−Re and Nd−Ru in the first, Al−Mg and C−Mg in the third region. These boundaries are characterized by a dominance of nonforming and ordered phases. The ordered phases form through crystallization from an expanded solubility range of their constituent chemical elements, and their ordered crystal

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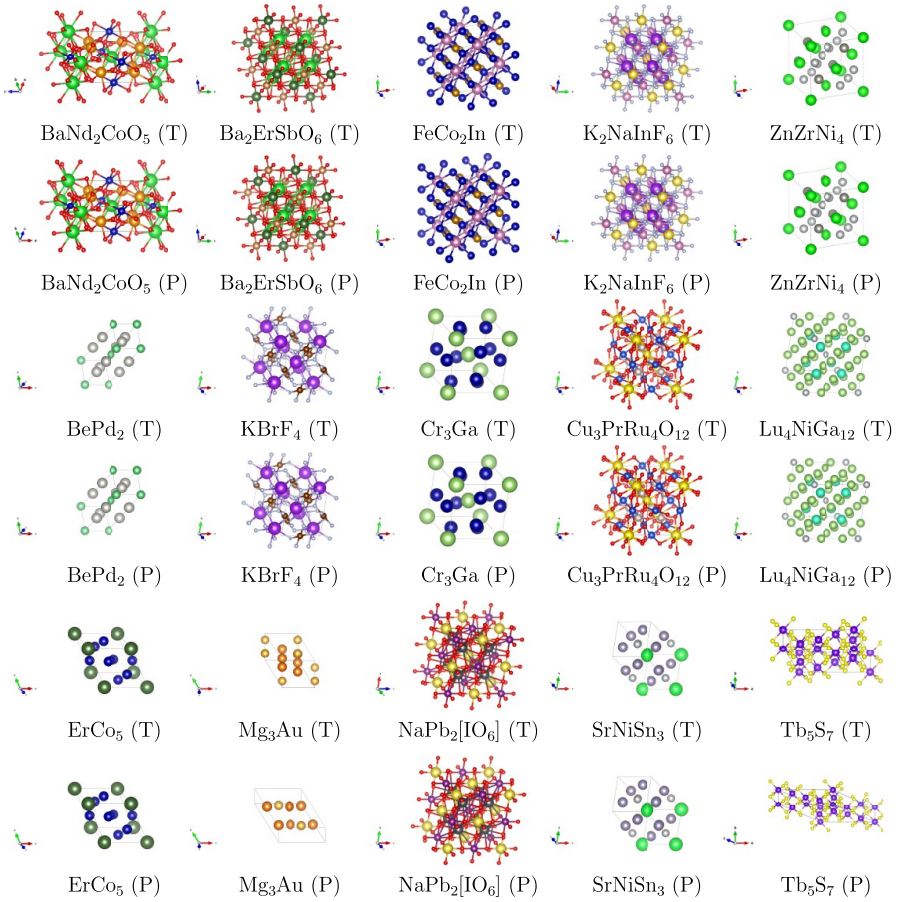


Figure 5. Target (T) and predicted (P) crystal structures of 15 phases taken from the crystal structure prediction benchmark set.[31]

structures are represented by the chemical element prototypes of these constituents, as highlighted in.[22] Mo−Sn, S−Se and Sb−Sn, despite not directly bordering a forbidden region, share this characteristic environment.

In the phase diagram, the equimolar phase of Bi−Ir is indicated for temperatures greater than 470 K. Our calculations reveal that the Bi−Ir equimolar phase, which adopts the NiAs,hP4,194 structure type, exhibits a positive enthalpy of formation at 0 K. Despite this, it is both mechanically and dynamically stable. Furthermore, for this phase, the Gibbs free energy becomes negative with increasing temperature, shifting from 25.5 kJ/mol at 298 K to −8.8 kJ/mol at 600 K, and reaching −65.8 kJ/mol at 1000 K. Consequently, these findings strongly support the conclusion that it is a high-temperature phase.

The equimolar phase of C−Yb is quite particular. In fact, though the system is surrounded by neighbors with distinct stable modifications, like for example the orthorhombic modification of YbSi, none of the neighboring prototypes resulted in fully stable equimolar phase of C−Yb at T = 0 K, except the rock-salt structure, as the phonon calculations and the derived thermodynamic properties suggest, which is most likely a high-temperature modification. The calculated enthalpy of formation at T = 0 K is positive, 78.06 kJ/mol, but only at temperatures as high as 1000 K the Gibbs free energy becomes 7.91 kJ/mol. Similarly, the enthalpy of formation of YC at T = 0 K is slightly negative, −4.49 kJ/mol in the present calculations, but the Gibbs free energy, derived from the phonon calculations is −14.06 kJ/mol at 500 K. This is not surprising, as the binary monocarbides of RE elements are nonformers in an exact equimolar composition and they are generally high-temperature phases; however, to the best of our knowledge, they are not extensively reported in the literature. Our calculations show that the thermodynamic stability of the rock-salt modification of the stoichiometric monocarbides of RE elements Sc, Y, La and Ce decrease along the group. In addition, the behavior of heavy RE monocarbides is reported in the literature as being different from the light RE monocarbides. In fact, our calculations suggest that while the cubic structure of ScC is completely stable, with a negative enthalpy of formation of −42.07 kJ/mol, and mechanical and dynamical stability, the stability of YC decreases (enthalpy of formation slightly negative −4.49 kJ/mol) and the trend is followed by LaC (92.07 kJ/mol) and CeC (−9.63 kJ/mol) up to YbC with +78.06 kJ/mol. Nevertheless, our results are in agreement with earlier studies, where the Yb-monocarbide was reported to form most likely either in the nonstoichiometric phase or not in a single phase.[30] In general, the enthalpies of formation of the identified thermodynamically stable structures align well with the trends observed in the binary phase map.[22]

In Figure 3, we report the phase map in the PN representation of the newly found equimolar phases and in Figure 4, we incorporate the corresponding values of the enthalpies of formation of these new phases in the already reported values of the equimolar binary phases.[22] From the heat map, we can see that the stability trends are satisfied by the new modifications, reinforcing once more the importance of the PN as a qualified descriptor to catch and predict

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Table 2. Our Predicted Structures of 15 Target Phases of Binary, Ternary, and Quaternary Systems by Using the PN Similarity Method[a]

reference system	system category	target phase	structure type (predicted)	$\Delta_f H(0)$ [eV/atom]	similarity score
mp-1231	binary_easy	Cr_3Ga	Cr_3Si,cP8,223	−0.090	0.00
mp-2740	binary_medium	$ErCo_5$	$CaCu_5$,hP6,191	−0.057	0.03
mp-13452	binary_hard	$BePd_2$	$MoSi_2$,tI6,139	−0.527	0.05
mp-11251	polymorph_binary	Mg_3Au	Mg_3Cd,hP8,194	−0.347	2.80
mp-1095641	template-based_binary	Tb_5S_7	Y_5S_7,mS24,12	−2.142	0.05
mp-21211	ternary_easy	$FeCo_2In$	Cu_2MnAl,cF16,225	0.083	0.00
mp-30805	ternary_medium	$SrNiSn_3$	$BaNiSn_3$,tI10,107	−0.502	0.12
mp-11533	ternary_hard	$ZnZrNi_4$	$MgCu_4Sn$,cF24,216	−0.473	0.00
mp-23550	polymorph_ternary	$KBrF_4$	$KBrF_4$-b,tI24,140	−2.006	0.12
mp-1106196	template-based_ternary	Lu_4NiGa_{12}	Y_4PdGa_{12},cI34,229	−0.551	0.02
mp-6686	quaternary_easy	K_2NaInF_6	Ba_2CaWO_6,cF40,225	−3.177	0.02
mp-6231	quaternary_medium	Ba_2ErSbO_6	Ba_2CaWO_6,cF40,225	−2.847	0.02
mp-19118	quaternary_hard	$BaNd_2CoO_5$	$BaNd_2NiO_5$,oI18,71	−2.724	0.18
mp-1105109	template-based_quartenary	$Cu_3PrRu_4O_{12}$	$CaCu_3Mn_4O_{12}$,cI40,204	−1.397	0.02
mp-1080753	polymorph_quartenary	$NaPb_2[IO_6]$	Ba_2CaWO_6,cF40,225	−0.999	0.16

[a]For the sake of easy comparison with the enthalpies of formation reported for the reference systems, we used here the units of eV/atom.

property trends. In fact, as expected, modifications with positive formation energies (represented by empty cells with borders) are predominantly situated near the nonforming and ordered phase regions in the PN-map. The key finding here is the compatibility between the stability of identified structures (negative enthalpy of formation) and the overall trends in the phase map. This suggests that the method successfully identified stable materials within regions typically associated with stable phases, while those with positive formation energies (unstable) tend to cluster near regions known for being nonforming or ordered phases.

The trend highlighted by the PN representation was also found in the structure types adopted by the stable modifications along a given group. For example, for the binary stannides, around 31 systems are reported, to our best knowledge, as having the equimolar phase either at room temperature or high pressure. Using the PN representation and the concept of crystal structure similarity, we identified additional monostannides, such as CeSn, TiSn, and YSn, which are fully stable with the respective enthalpies of formation at T = 0 K: CeSn (TlI,oS8,63) −120.30 kJ/mol; TiSn (FeSi,cP8,198) −62.46 kJ/mol; YSn (CsCl,cP2,221) −121.52 kJ/mol.

Another significant point is that nearly half of the identified stable prototypes represent entirely novel polymorphic phases, absent from existing open-access databases (the complete list is presented in Table 1 of the SI document). Moreover, of these 34 'new' polymorphic phases, 19 are fully stable. The remaining 61 predicted polymorphs align with structures previously proposed in various studies and documented within OQMD. This result highlights two key points. First, the method successfully identifies a substantial number of previously unknown stable materials, locating their stable phases precisely in the PN representation in a knowledge-based way. This advancement expands the known repertoire of potentially useful materials. Second, the identification of modifications in agreement with the existing data in OQMD validates the method's capability of predicting stable polymorphs with the lowest energy values. This suggests our method has the potential to predict the most thermodynamically stable structures.

Beyond this, PNscp is designed to perform crystal structure prediction not only for equimolar binary systems but also any phases of higher order chemical systems. In this respect, we extended our target set to encompass binary, ternary, and quaternary systems with varying stoichiometry, chemical compositions and hence chemical interactions.

To rigorously assess the algorithm's performance, we selected 15 representative phases, which are shown in Figure 5, taken from the crystal structure prediction benchmark set.[31] According to the developers of the benchmark set, the data set, collected from Materials Project,[2] includes structures with varying levels of prediction difficulty. Therein the authors use the criteria for classifying difficulty encompassing factors such as space group classification, template-based categorization and prototype ratios that define crystal structures. To test our algorithm, we picked up a range of cases balanced by difficulty —easy, medium, hard, polymorph, and template-based— as categorized by the authors.[31] Doing that we ensured a thorough evaluation of the algorithm's efficacy across varying levels of complexity.

During the test, we used OQMD as the primary source for gathering neighboring structures. Given that the queries in the OQMD searching often yield a large number of redundant candidate structures, we implemented a strategy to mitigate this redundancy. We identified and removed duplicate prototypes by retaining only one prototype from a set of highly similar structures. This approach effectively reduces the number of duplicates and indistinguishable prototypes for the same phase. Following this reduction, geometry optimizations were performed on the selected prototypes.

The structural similarity between target and predicted structures was assessed using the CrystalNN function from the Matminer package.[32] This measure incorporates features from both the site of interest and its neighboring sites in a vector encapsulating coordination environment details. By statistical analysis of these site features, a structural fingerprint is generated for each structure. The structural similarity index of the target and predicted structures is then quantified by calculating the distance between their respective fingerprint vectors. A small distance value represents a high degree of similarity between two structures, while a large distance (>1) indicates a dissimilarity between them.[33]

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In Table 2, we list 15 selected chemical systems along with their structure types of the optimized structures, their respective enthalpies of formation and similarity score which indicates how much the predicted structures differ from the target ones. The crystallographic images of the reference structures for target systems as well as our predicted structures are presented in Figure 5. The results demonstrate that our algorithm successfully identifies the experimental phases for almost all the considered systems, thereby affirming its ability to manage diverse chemical compositions effectively.

The Mg_3Au system is a notable exception, exhibiting a significant deviation from the experimental structure. Even though the space group of the predicted crystal structure aligns with the target, there are certain structural differences between them, leading to a high similarity score of 2.8. Furthermore, it is noteworthy that the predicted structure for Mg_3Au has a negative enthalpy of formation. Actually, the Au−Mg system, which has been studied since 1930, presents several intermetallic phases, among which three phases were reported in literature as stoichiometric single phases: MgAu, Mg_2Au and Mg_3Au with structure types of CsCl,cP2,221, Mg_2Au,oP108,62, Na_3As,hP8,194, respectively.[30] The latter phase, which we picked up for testing our similarity algorithm, was reported in the complete experimental structure determination with a structure type different from that assigned in the reference system on Table 2. The structure type at phase level is Cu_3P,hP24,185 though there are in the literature other experimental studies proposing the structure type Na_3As,hP8,194 for thin films, and HoH_3,hP24,165 for powder samples.

CONCLUSIONS

This work presents a novel approach to search and predict prototypes of materials leveraging fundamental chemical similarities encoded within Mendeleev PN. The proposed similarity method, PNcsp, demonstrates notable efficiency in proposing crystalline structures of target chemical systems for which a limited extent of empirical knowledge is available. Its success highlights the concept of the PN of elements for finding similarity in chemical systems in any constitution. Though the current study mainly focus on equimolar binary systems, we demonstrated that the method can be readily applied to ternary, quaternary, and higher order chemical systems, facilitating the discovery of novel crystalline phases within the phase space.

The core advantage of this methodology lies in its simplicity and the fact that it is based on the chemical element periodicity. Unlike complex ML models, the PN-based similarity approach provides a clear picture of the underlying factors contributing to phase stability. The prediction accuracy of the method, in comparison to the purely data-driven approaches, lies in the narrowing the phase space to span for searching for stable phases, with the advantage of reducing the computational cost but also helping to pick up the most chemically reasonable modifications. In addition, this approach not only enhances the efficiency of the structure prediction process but also facilitates the discovery of previously unreported polymorphs. Our findings demonstrate the potential of leveraging the intrinsic logic of the periodic table to enhance innovation and discovery in materials science.

ASSOCIATED CONTENT

Data Availability Statement

An open-source software implementation of our similarity-based initial crystal structure prediction method is available at https://github.com/tccdem/PNcsp.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.inorgchem.4c03137.

A summary of all modifications exhibiting a negative enthalpy of formation and mechanical properties and elastic constants of fully stable modifications are provided (PDF)

Crystal structures of fully stable modifications and CSPBenchmark (ZIP archive)

AUTHOR INFORMATION

Corresponding Authors

Riccarda Caputo − *Computational Materials Informatics - CMI, 00157 Rome, Italy*; Email: riccarda.caputo@outlook.com

Pierre Villars − *Material Phases Data System (MPDS), 6354 Vitznau, Switzerland*; Email: villars.mpds@bluewin.ch

Adem Tekin − *Informatics Institute, Istanbul Technical University, 34469 Istanbul, Türkiye; TÜBİTAK Research Institute for Fundamental Sciences, 41470 Gebze, Türkiye*; orcid.org/0000-0002-0123-4197; Email: adem.tekin@itu.edu.tr

Authors

Cem Oran − *Informatics Institute, Istanbul Technical University, 34469 Istanbul, Türkiye*

Hasan Bilal Özcü − *Informatics Institute, Istanbul Technical University, 34469 Istanbul, Türkiye*

Feraye Hatice Canbaz − *Informatics Institute, Istanbul Technical University, 34469 Istanbul, Türkiye*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.inorgchem.4c03137

Author Contributions

C.O.: Methodology, Data curation, Writing−original draft, Writing−review and editing, Visualization. **R.C.:** Conceptualization, Methodology, Data curation, Supervision, Writing−original draft, Writing−review and editing. **P.V.:** Conceptualization, Writing−review and editing. **H.B.Ö.:** Data curation. **F.H.C.:** Data curation. **A.T.:** Supervision, Data curation, Writing - review and editing.

Notes

The authors declare no competing financial interest.

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